ARISTOTLE UNIVERSITY
OF THESSALONIKI

R data wrangling

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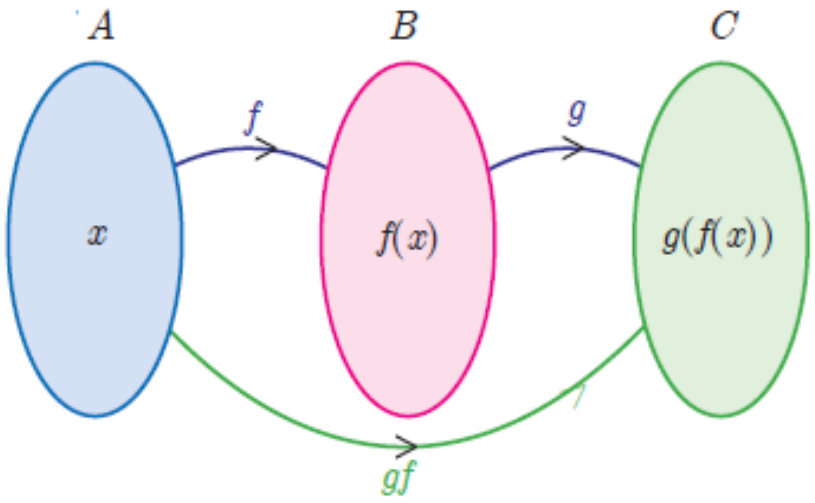
THESSALONIKI

Core tidyverse packages

Pipe operator |> (or %>%)

Perform a sequence of operations on a data frame x using the functions f(), and g() :

```
g(f(x))
```



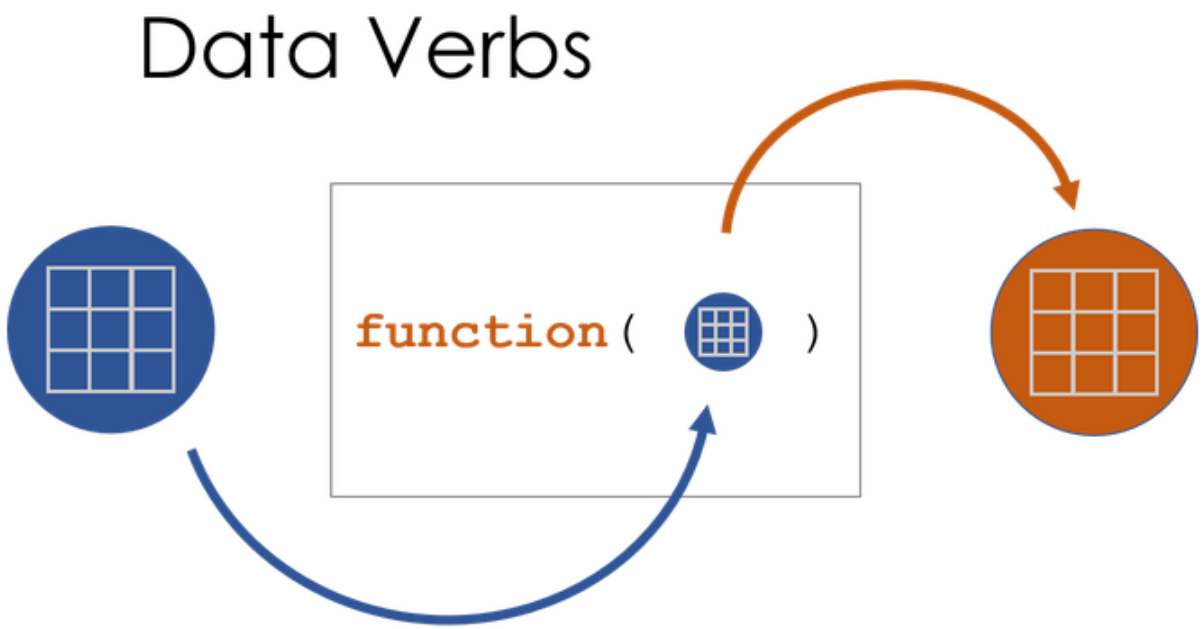
With |>:

```
x |>
  f() |>
  g()
```

We would read this sequence as:

1. Take x *then*
2. Use this output as the input to the function f() *then*
3. Use this output as the input to the next function g()

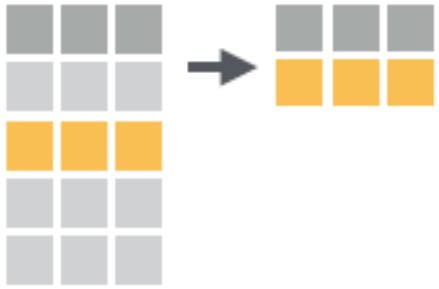
Common dplyr functions



A data verb requires a dataset as input, and returns a transformed dataset.

Common dplyr functions: filter() and select()

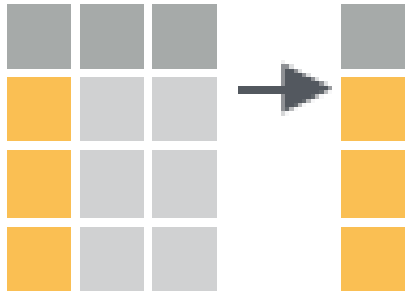
EXTRACT CASES



`filter()`

Extract rows that meet logical criteria.

EXTRACT VARIABLES

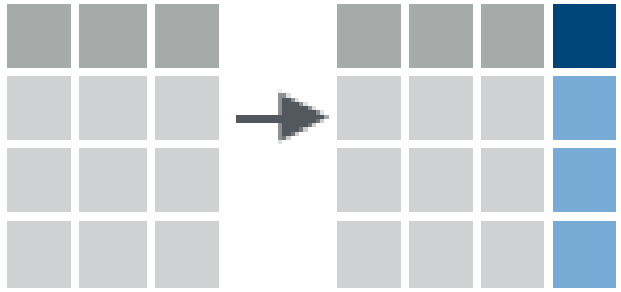


`select()`

Extract columns as a table.

Common dplyr functions: mutate() and group_by()

MAKE NEW VARIABLES



mutate()

Creates a new transformed variable from the formula we specify and adds it to the end of the original dataset.

GROUP THE DATA

city	particle size	amount ($\mu g/m^3$)
New York	large	23
New York	small	14
London	large	22
London	small	16
Beijing	large	121
Beijing	small	56

city	particle size	amount ($\mu g/m^3$)
New York	large	23
New York	small	14
London	large	22
London	small	16
Beijing	large	121
Beijing	small	56

group_by()

It takes an existing data table and converts it into a grouped data table where operations are performed "by group“.